An Item-Level Audit Protocol for Error Persistence and Regression Burden in Small Language Model Replacement

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Aggregate benchmark accuracy is insufficient on its own for replacing a deployed small language model. A candidate model can improve the mean score while newly failing items that the current model answered correctly. This paper treats replacement as a paired item-level audit: each shared labeled item is assigned to persistent correct, corrected error, regression, or persistent error, yielding correction rate, regression mass, churn mass, error persistence, and normalized regression burden.

The contribution is a practical audit protocol and artifact, not a scaling law or a causal account of model families. A public WILD correctness-only demonstration retains 276,685 item-level records across five Qwen and Llama models up to approximately four billion parameters and ten tasks. The Qwen2.5-0.5B to Qwen2.5-3B comparison improves aggregate correctness by 22.2 percentage points while changing 40.9% of item outcomes; the near-parity Qwen2.5-1.5B to Qwen2.5-3B comparison improves by 5.7 points while changing 30.4% of outcomes. An archived raw-output MMLU-Pro case study covers 14 model-output runs over 3,008 reconstructed and hashed items, but remains diagnostic because the original logs lack complete inference manifests and the manual parser audit is incomplete. The operational recommendation is direct: benchmark-based replacement decisions should report corrected errors, regressions, persistent errors, and churn, not aggregate accuracy alone.

1 Introduction

Models up to approximately four billion parameters are often considered for deployment because they can run under tighter memory, latency, privacy, and cost constraints than larger frontier systems. This paper uses that evaluated size range as its operational scope because it matches the retained model set and the deployment regime where local memory and latency constraints commonly motivate replacement decisions. In such settings, replacement is a common engineering decision. A practitioner has a current model, evaluates a candidate model checkpoint or larger variant, and may adopt the candidate model if its aggregate benchmark accuracy is higher.

That decision rule is incomplete. A replacement can fix old errors while breaking items that the current model handled correctly. In a document triage system, for example, a small aggregate improvement may still introduce new failures on previously reliable cases. The replacement may still be worth adopting, but the tradeoff should be visible before deployment.

This paper reframes replacement as a paired item-level audit. Given two models evaluated on the same labeled items, we separate persistent correct items, corrected errors, regressions, and persistent errors. This decomposition measures what aggregate accuracy hides: which errors remain, which errors are fixed, which old successes are newly broken, and how much of the item-level correctness pattern changes.

The central question is deliberately narrow:

> When a small language model improves in aggregate, how much of the item-level correctness pattern actually changes?

The paper makes four contributions.

(1) It gives a reusable replacement-audit protocol based on paired item-level correctness, including correction rate, regression mass, churn mass, error persistence, and normalized regression burden.
(2) It separates deployment-path audits from standardized all-pairs diagnostics. Directional metrics are meaningful only with respect to a specified current model and candidate model.
(3) It demonstrates the audit on a public WILD correctness-only matrix, so the core accounting result does not depend on private raw model completions or answer parsing.

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(4) It provides a diagnostic archived-output MMLU-Pro case study with reconstructed source-item hashes, scoring-mode sensitivity checks, and explicit limitations on answer extraction and run provenance.

The claim is not that prediction churn is new, that Qwen or Gemma has a generalizable family-level property, or that the reported numbers are a scaling law. The claim is practical: if benchmark scores are used to justify replacing a model, then corrected errors and newly introduced regressions should be reported alongside the aggregate score.

2 Related Work

Beyond aggregate accuracy. NLP evaluation has repeatedly shown that aggregate scores can hide important behavior. CheckList introduced behavioral tests to expose failures hidden by held-out accuracy (Ribeiro et al. 2020). HELM argues for transparent, multi-metric model evaluation rather than single-number comparisons (Liang et al. 2022). The present audit follows the same evaluation philosophy, but focuses on paired item-level changes between a current model and a candidate replacement model.

Prediction churn and regression bugs. Prediction churn studies how model predictions change across retraining or deployment iterations (Fard et al. 2016). Regression-bug work in NLP model updates measures negative flips: cases where a newer model newly fails examples an older model handled correctly (Xie et al. 2021). Unlike work that proposes training or update methods to reduce regressions, this paper proposes a reporting and audit artifact for replacement decisions. The present work does not claim to invent churn or regression measurement. Its narrower contribution is to adapt this accounting to small language model replacement and to keep beneficial churn, harmful churn, and persistent old errors separate in the artifact and reporting.

Regression testing and deployment audit. The audit is close in spirit to software regression testing. Before replacing a component, users need to know which previously passing cases now fail. In model evaluation, the analogous question is which items were previously correct and are now wrong. This is a deployment-facing question rather than a claim about training mechanisms.

Production readiness and documentation. Production ML work emphasizes systematic testing, monitoring, and documentation rather than isolated aggregate scores (Breck et al. 2017; Sculley et al. 2015). Dataset and model documentation frameworks likewise stress provenance, intended use, and evaluation context (Gebru et al. 2021; Mitchell et al. 2019). The present artifact follows that orientation by separating source provenance, evidence tier, scoring status, and claim use.

Stability, forgetting, and underspecification. The replacement problem also connects to work on catastrophic forgetting (Kirkpatrick et al. 2017) and underspecification (D'Amour et al. 2022). Those literatures study different mechanisms and failure modes. Here, the object is narrower: observable item-level correctness changes between archived model outputs or public correctness records.

Benchmarks and public item-level resources. Benchmark suites such as MMLU, MMLU-Pro, and BIG-bench provide broad task-level measurement (Hendrycks et al. 2021; Srivastava et al. 2022; Wang et al. 2024). WILD provides public item-level correctness records that can be used for correctness-only replacement audits (Krumdick et al. 2026). Aggregate leaderboards and model cards are useful for context, but they cannot identify which old errors were corrected or which previously correct items regressed.

Tier	Contents	Permitted use
A	Raw prompts, raw completions, labels, item IDs, and validated scoring	Full audit, parser analysis, format-failure analysis
B	Item IDs, model IDs, and binary correctness	Correctness-only churn, regression, persistence, and correction metrics
C	Aggregate benchmark scores only	Context only; no item-level replacement claims
D	Gated, unavailable, incompatible, or unverified data	Excluded until access and terms are clear

Table 1. Evidence tiers for replacement accounting. Only Tier A and Tier B sources support item-level churn metrics.

Evidence component	Status	Claim use
WILD item-level correctness	Supported	Core replacement-accounting evidence for persistence, regression mass, and churn over paired correctness records.
MMLU-Pro source reconstruction	Supported	Provenance evidence that all 3,008 evaluated item identifiers map to the stated MMLU-Pro snapshot.
MMLU-Pro raw-output accounting	Diagnostic only	Demonstrates the raw-output workflow and artifact-risk checks, but remains parser-dependent.
Parser-validation claims	Not claimed	Human labels are incomplete, so the paper does not report parser agreement as validation.
Aggregate benchmark context	Context only	Positions model capability; does not support item-level replacement or churn claims.

Table 2. Claim ledger for the current GitHub/SSRN release. Parser-dependent MMLU-Pro claims are explicitly excluded from the core claim set until human validation is complete.

3 Replacement-Audit Protocol

3.1 Inputs and evidence tiers

The audit requires paired item-level outcomes for two models on the same labeled item identifiers. The strongest input is raw model output plus a validated scoring procedure, meaning either benchmark-provided predictions or a completed human-labeled parser audit. A weaker but still useful input is a public binary correctness matrix. Aggregate leaderboards alone are insufficient.

3.2 Four-cell decomposition

For a directed comparison from current model A to candidate model B over the same N items, each item belongs to exactly one cell:

- persistent correct (PC): both models are correct;
- improvement or corrected error (I): A is wrong and B is correct;
- regression (R): A is correct and B is wrong;
- persistent error (PE): both models are wrong.

The aggregate accuracy delta is

$$\Delta = \frac{|\mathrm{I}| - |\mathrm{R}|}{N}. \tag{1}$$

The audit metrics are

$$\text{Improvement mass} = \frac{|\mathrm{I}|}{N}, \tag{2}$$

$$\text{Regression mass} = \frac{|\mathrm{R}|}{N}, \tag{3}$$

$$\text{Churn mass} = \frac{|\mathrm{I}| + |\mathrm{R}|}{N}, \tag{4}$$

$$\text{Error persistence} = \frac{|\mathrm{PE}|}{|\mathrm{PE}| + |\mathrm{I}|}, \tag{5}$$

$$\text{Correction rate} = \frac{|\mathrm{I}|}{|\mathrm{PE}| + |\mathrm{I}|}. \tag{6}$$

Let the current model's error rate be

$$e_0 = \frac{|\mathrm{PE}| + |\mathrm{I}|}{N}. \tag{7}$$

The normalized regression burden is

$$\mathrm{NRB} = \frac{|\mathrm{R}|/N}{e_0}. \tag{8}$$

If $|\mathrm{PE}| + |\mathrm{I}| = 0$, the current model has no observed errors on the shared item set. Error persistence, correction rate, and NRB are then undefined and are reported as `NA`, not as zero. Category summaries with zero current-model errors remain included for mass metrics such as improvement mass, regression mass, and churn mass, but old-error-normalized ratios are reported as `NA`.

For an accuracy-improving directed comparison, NRB is a practical normalization of regression mass by the current model's observed error rate. It can also be read descriptively as the gap between the no-regression upper bound on old-error persistence and observed old-error persistence. This interpretation is descriptive; it is not a theoretical novelty claim and does not identify a causal mechanism.

3.3 Directed paths versus all-pairs diagnostics

The metrics are directional. A comparison $A \longrightarrow B$ answers a different deployment question than $B \longrightarrow A$. A deployment audit should therefore use the actual proposed replacement path whenever it is known.

When only a set of model outputs is available, a standardized all-pairs diagnostic can still be useful. In that case, the ordering rule must be explicit and should not be mistaken for vendor release chronology. In the MMLU-Pro case study below, directed pairs are ordered by parameter count, then generation string, then model label. Comparisons marked as chronology unclear are treated as diagnostic pairs, not as true deployment histories.

3.4 Minimal audit-card algorithm

(1) Align current-model and candidate-model records by item identifier.
(2) Convert each model's output to a binary correctness label under a documented scoring procedure.
(3) Count PC, I, R, and PE.
(4) Compute Δ, improvement mass, regression mass, churn mass, error persistence, correction rate, and NRB.
(5) Emit an audit card with optional review flags for high churn, high regression mass, high NRB, near-parity churn, and negative-delta swaps.

3.5 Statistical posture

The audit is primarily descriptive. McNemar-style paired tests are appropriate when comparing two classifiers on the same items (Dietterich 1998; McNemar 1947), but replacement accounting should not treat shared-endpoint

Field	Purpose
Current and candidate model accuracy	Aggregate movement
Accuracy delta	Net gain or loss
Improvement mass	Share of all evaluated items that the current model missed and the candidate model corrected
Regression mass	Share of all evaluated items that the current model answered correctly and the candidate model newly misses
Churn mass	Total share of changed item outcomes
Error persistence and correction rate	How much of the current error set remains
Normalized regression burden	Practical normalization of regression mass relative to current-model errors
Top regressing categories or clusters	Where manual review should start
Path note	Whether the directed pair is a real replacement path or a diagnostic comparison

Table 3. Minimal replacement-audit card emitted by the artifact.

model pairs as independent evidence. The artifact therefore reports mass and churn metrics as the primary outputs and uses inferential checks only as clearly labeled diagnostics.

4 Public WILD Correctness-Only Demonstration

The WILD demonstration is the parser-independent empirical support for the audit protocol because it avoids answer extraction. WILD exposes item identifiers, model identifiers, task identifiers, and binary correctness records. The full WILD release contains evaluations of 65 models over 109,564 unique items spanning 163 tasks from 27 datasets (Krumdick et al. 2026). The present artifact uses a selected public subset containing five Qwen and Llama models up to approximately four billion parameters and ten tasks. It supports item-level churn accounting, but it does not support raw-response parsing claims, prompt-format claims, or manual answer-extraction validation.

The WILD subset rule is fixed in the artifact: retain public item-level correctness rows for the selected Qwen2.5 chain and Llama 3.2 1B/3B models where the selected models share task coverage. The selected tasks are MMLU-Pro, MMLU, ARC-Challenge, GSM8K, HellaSwag, PIQA, BoolQ, IFEval, BBH, and MATH. This is a public cross-benchmark demonstration of the audit protocol, not a representative analysis of all WILD models or all WILD tasks.

The generated WILD replication contains 44 directed same-family comparisons: four selected-task pooled comparisons, plus 40 task-level comparisons across ten tasks. Across the selected WILD tasks, the three Qwen selected-task pooled directed comparisons have 34.4% mean churn. The Qwen2.5-0.5B to Qwen2.5-3B comparison improves aggregate correctness by 22.2 percentage points while changing 40.9% of item outcomes and regressing 9.3% of all shared items. The near-parity Qwen2.5-1.5B to Qwen2.5-3B comparison improves aggregate correctness by 5.7 percentage points while changing 30.4% of item outcomes. This is the core operational result in a parser-free setting: aggregate gain and item-level replacement churn are different quantities.

The Llama 3.2 1B to 3B comparison gives the same qualitative lesson. It improves by 17.8 percentage points while changing 35.3% of item outcomes and regressing 8.8% of all shared items. The audit therefore identifies not only whether a candidate model wins on average, but how much of the correctness pattern is rewritten.

The task-level dispersion table shows why an all-task mean is not enough. Across all selected WILD task-level comparisons, median churn is 33.9% with an interquartile range of 30.0–39.4%. For Qwen alone, median task-level churn is 32.6%, and the highest mean-churn task among the selected tasks is PIQA at 42.9%. These values do not

Model	Family	Size	Tasks	Included	Reason
Qwen1.5-4B-Chat	Qwen	4.0B	10	no	Different Qwen generation; not part of the predeclared Qwen2.5 replacement chain.
Qwen2-1.5B-Instruct	Qwen	1.5B	10	no	Single Qwen2 small model in WILD; no retained same-generation small replacement pair.
Qwen2.5-0.5B-Instruct	Qwen	0.5B	10	yes	Retained in the predeclared Qwen2.5 size chain.
Qwen2.5-1.5B-Instruct	Qwen	1.5B	10	yes	Retained in the predeclared Qwen2.5 size chain.
Qwen2.5-3B-Instruct	Qwen	3.0B	10	yes	Retained in the predeclared Qwen2.5 size chain.
granite-3.2-2b-instruct	Granite	2.0B	10	no	No second under-4B Granite model in WILD for a same-family replacement pair.
llama-3.2-1b	Llama	1.0B	10	yes	Retained in the Llama 3.2 small-model replacement pair.
llama-3.2-3b	Llama	3.0B	10	yes	Retained in the Llama 3.2 small-model replacement pair.
Nemotron-Mini-4B-Instruct	Nemotron	4.0B	10	no	No second under-4B Nemotron model in WILD for a same-family replacement pair.

Table 4. WILD inclusion coverage for recognizable public WILD models at approximately four billion parameters or below. The table reports the selected-task rows available in WILD and the reason each model was retained or excluded from the paired-family demonstration.

Field	Value
External source	WILD
Evidence tier	B: item-level correctness
Models retained	5
Tasks retained	10
Normalized records	276685
Pairwise comparisons	44 (4 selected-task pooled + 40 task-level)
Selected-task pooled Qwen mean churn	34.4%
WILD source scale	65 models; 109,564 items; 163 tasks; 27 datasets

Table 5. Public WILD correctness-only demonstration.

Family	Current	Candidate	N	Δ	I	R	Churn	Persist.	Corr.	NRB
llama	llama-3.2-1b	llama-3.2-3b	54463	+17.8%	26.5%	8.8%	35.3%	58.1%	41.9%	13.9%
qwen	Qwen2.5-0.5B-Instruct	Qwen2.5-1.5B-Instruct	54464	+16.5%	24.2%	7.7%	31.9%	64.2%	35.8%	11.4%
qwen	Qwen2.5-0.5B-Instruct	Qwen2.5-3B-Instruct	54464	+22.2%	31.5%	9.3%	40.9%	53.3%	46.7%	13.8%
qwen	Qwen2.5-1.5B-Instruct	Qwen2.5-3B-Instruct	54464	+5.7%	18.0%	12.3%	30.4%	64.7%	35.3%	24.2%

Table 6. Selected-task pooled WILD directed comparisons for retained small Qwen and Llama models. These are correctness-only item-level replacement metrics, not parser or raw-response analyses. I is improvement mass, R is regression mass, and NRB is normalized regression burden.

imply a universal churn level for all small models. They show that the accounting method can be applied to a public item-level correctness matrix and that large churn appears outside the MMLU-Pro archived-output case study.

Scope	Tasks	Pairs	Median N	Median churn	Churn IQR	Median R	Median persist.	Highest-churn task
selected-task pooled	10	40	4135	33.9%	[30.0%, 39.4%]	7.4%	47.9%	piqa (41.8%)
llama	10	10	4135	34.8%	[32.8%, 37.8%]	6.9%	45.6%	hellaswag (43.7%)
qwen	10	30	4135	32.6%	[28.0%, 39.8%]	7.4%	50.2%	piqa (42.9%)

Table 7. Task-level dispersion for WILD directed comparisons. The table summarizes the 40 task-level comparisons, separate from the four selected-task pooled comparisons in Table 6, and reports shared-item denominators.

Field	Value
Evaluated items	3008
Matched MMLU-Pro rows	3008
Mismatches	0
Missing source rows	0
Manifest reconstruction passed	yes

Table 8. MMLU-Pro source-manifest reconstruction. The public artifact stores row hashes and source identifiers, not benchmark question text or answer-option text.

5 MMLU-Pro Archived-Output Case Study

5.1 Source manifest

The MMLU-Pro case study uses 14 archived model-output runs over 3,008 shared items. A source-manifest script reconstructs the evaluated rows from the TIGER-Lab/MMLU-Pro test parquet snapshot and verifies saved identifiers against source category and answer labels. The manifest stores normalized item hashes and source identifiers, not benchmark question text. The reconstructed source revision is b189ec765aa7ed75c8acfea42df31fdae71f97be; the source parquet SHA256 is 0e24a191921c2f453518a537a8b2117bd137e7714d4ef1565e9ba06c1ecb9ad8; and the public manifest item-hash SHA256 is 1f482b880fad0d8c97aa08f83d250ffd43e715b78ec2619987e0db4e9a98d9d3.

All 3,008 evaluated identifiers match the source rows with zero category or answer mismatches. The subset is now reconstructable, but the original reason for selecting exactly these 3,008 rows from MMLU-Pro was not preserved. The MMLU-Pro case study should therefore be read as an audit of a fixed archived subset, not as a statistically designed sample of the whole benchmark.

5.2 Answer extraction and scoring validation

The MMLU-Pro raw outputs are not perfectly uniform. Three runs include saved prediction fields; the remaining rows require deterministic extraction from raw responses. Each row is scored as answered or unanswered. A conservative prompt-echo guard marks responses ending in a bare chat marker after echoed prompt text as unanswered. This matters for Gemma-4-E2B-Instruct: after the guard, its answered rate is 5.2% and parse-failure rate is 94.8%. That run is retained in descriptive files as a generation-format failure but excluded from primary pairwise claims.

A manual parser-audit package has been generated, including both family/method stratified rows and high-risk rows selected from fallback methods, prompt echoes, low-confidence parser decisions, and saved/raw disagreement cases. Human labels are not complete. The present paper therefore does not report human/parser agreement and does not present the MMLU-Pro raw-output case study as a fully validated model evaluation.

Family	Models	Rows	Saved pred.	Explicit marker	Fallback	Unanswered	Saved-field models
Gemma	5	15040	5855	4169	1929	3087	2
Llama	1	3008	0	2440	513	55	0
Phi	1	3008	2730	0	0	278	1
Qwen	7	21056	0	15282	5686	88	0

Table 9. Answer-source coverage by family. Saved predictions are used when present; otherwise explicit answer markers and fallback letter rules are applied to raw responses.

Validation item	Value
Audit sample rows	452
High-risk audit rows	120
Completed human-labeled rows	0
Saved-prediction consistency rows	8585
High-confidence sensitivity families	4
Manual audit ready	no

Table 10. Parser-validation status for the archived MMLU-Pro raw-output case study. Manual labels are required before reporting parser agreement as a validation result.

Dimension	Value	Rows	Agree	False corr.	False incorr.	False unans.
overall	pending labels	0	–	–	–	–
Claim gate	blocked	0	–	0	0	–

Table 11. Parser-audit impact gate. The current artifact has generated the sample but has not completed human labels, so MMLU-Pro remains a parser-dependent diagnostic.

The parser-audit protocol is preregistered in the artifact. The human labeler sees the source item identifier, model/family, extraction method, audit source, risk reason, parser prediction, parser answered flag, ground-truth option letter, and a private raw-response excerpt or source response field. Benchmark question text and response excerpts are not distributed in the public release. The labeler fills `human_prediction`, `human_answered`, `human_parser_correct`, and notes. Multiple-answer cases are labeled by the answer a reasonable benchmark scorer would credit; ambiguous cases are marked in notes and adjudicated before final reporting; no-answer or prompt-echo cases are labeled unanswered unless a clear option letter is actually supplied. The gate requires at least 452 completed labels, including all 120 high-risk rows, and reports agreement overall, by family, by extraction method, and by audit source.

The parser-impact report extends the audit from extraction accuracy to claim impact. It counts false correct, false incorrect, false unanswered, false regression-impact, and false improvement-impact cases. With the current unlabeled package, Table 11 correctly blocks strong MMLU-Pro claims rather than imputing validation.

The scoring-mode sensitivity check uses the three runs that contain both saved predictions and raw responses. Table 12 compares the default saved-prediction-primary scoring with a raw-response-only parser. It shows that

Model	Saved acc.	Raw-only acc.	Δ	Saved answered	Raw answered	Changed corr.
Gemma-3-1B-Instruct	11.8	13.1	+1.4	99.1	99.0	11.6
Gemma-3-270M-Instruct	10.4	10.0	-0.4	95.5	95.5	3.3
Phi-2	11.8	10.9	-0.9	90.8	61.7	8.2

Table 12. Scoring-mode sensitivity for runs with both saved predictions and raw responses. Values are percentages. Δ is raw-response-only accuracy minus saved-prediction-primary accuracy.

Mode	Family	Pairs	Mean N	Δ	R	Churn	Persist.
default	gemma	10	3008	+0.5	11.8	24.1	85.9
default	qwen	21	3008	+17.2	8.9	35.0	65.8
exclude_low_confidence_fallback	gemma	10	1633	+22.8	6.8	36.4	65.5
exclude_low_confidence_fallback	qwen	21	1747	+23.8	7.6	39.1	54.7
exclude_saved_prediction_models	gemma	3	3008	-10.5	19.0	27.6	89.5
exclude_saved_prediction_models	qwen	21	3008	+17.2	8.9	35.0	65.8
explicit_marker_only	gemma	10	1633	+22.8	6.8	36.4	65.5
explicit_marker_only	qwen	21	1747	+23.8	7.6	39.1	54.7
raw_only	gemma	10	3008	+0.4	11.8	23.9	86.0
raw_only	qwen	21	3008	+17.2	8.9	35.0	65.8

Table 13. MMLU-Pro scoring-robustness modes over archived outputs. These are sensitivity analyses, not confirmatory validation while parser labels are incomplete.

scoring asymmetry is materially important, not merely measurable: Gemma-3-1B-Instruct changes correctness on 11.6% of rows across the two modes, and Phi-2's raw-response-only answerability is much lower than its saved-prediction-primary answerability. This sensitivity check is not a substitute for manual validation, but it prevents the paper from treating the parser as invisible.

The robustness generator also reruns MMLU-Pro accounting under five modes: default scoring, raw-only scoring where possible, exclusion of saved-prediction models, exclusion of low-confidence fallback extractions, and explicit-marker-only scoring where enough rows remain. The sign of the family contrast remains the same, but its magnitude changes under stricter extraction modes. The MMLU-Pro gate remains blocked because the manual parser audit is incomplete.

5.3 Run-provenance status

The archived raw run files preserve item IDs, categories, ground-truth labels, model labels, and raw responses. They do not preserve exact Hugging Face repository identifiers, revision hashes, prompt templates, chat templates, decoding parameters, runtime versions, run dates, or complete hardware metadata. Consequently, the MMLU-Pro accounting can be regenerated from saved outputs, but the original inference runs cannot be independently reproduced as validated model-behavior evaluations without rerunning or reconstructing the experiment.

Mode	Qwen pairs	Gemma pairs	Qwen persist.	Gemma persist.	Difference
default	21	10	65.8	85.9	-20.1
exclude_low_confidence_fallback	21	10	54.7	65.5	-10.8
exclude_saved_prediction_models	21	3	65.8	89.5	-23.7
explicit_marker_only	21	10	54.7	65.5	-10.8
raw_only	21	10	65.8	86.0	-20.2
Claim gate	blocked: manual parser audit is incomplete or below threshold				

Table 14. Family-contrast robustness gate for the archived MMLU-Pro diagnostic. The gate blocks confirmatory family language until manual parser validation passes.

Gate	Excluded models	Qwen pairs	Gemma pairs	Qwen persist.	Gemma persist.	Qwen NRB	Gemma NRB	Qwen churn	Gemma churn
none	none	21	10	65.8	85.9	12.8	14.5	35.0	24.1
70%	Gemma-4-E2B-Instruct	21	6	65.8	77.8	12.8	9.0	35.0	27.3
80%	Gemma-4-E2B-Instruct	21	6	65.8	77.8	12.8	9.0	35.0	27.3
90%	Gemma-4-E2B-Instruct	21	6	65.8	77.8	12.8	9.0	35.0	27.3

Table 15. Sensitivity of primary-family results to the answerability gate. Metrics are percentages. The no-gate row is descriptive only because it includes the Gemma-4-E2B-Instruct prompt-echo failure run.

Family	Models	Pairs	Old acc.	New acc.	Δ	I	R	Churn	Err. persist.	Corr. rate	NRB
Gemma	4	6	12.1	23.6	11.6	19.5	7.9	27.3	77.8	22.2	9.0
Qwen	7	21	24.3	41.4	17.2	26.1	8.9	35.0	65.8	34.2	12.8

Table 16. Primary MMLU-Pro same-family item-level accounting after excluding runs with answered rate below 80%. Values are percentages. I is improvement mass, R is regression mass, and NRB is normalized regression burden.

5.4 Primary population and quality gate

All 14 MMLU-Pro model-output runs and all 3,008 items are retained in descriptive outputs. A model is eligible for primary same-family pairwise claims if at least 80% of its rows have valid extracted answers. This gate excludes one run, Gemma-4-E2B-Instruct, whose outputs are dominated by prompt echoes rather than usable completions. The exclusion is treated as a generation-format incompatibility, not as a clean knowledge or reasoning failure.

After the gate, Qwen contributes seven eligible models and 21 same-family directed comparisons. Gemma contributes four eligible models and six same-family directed comparisons. Llama and Phi are retained as single-model anchors but excluded from same-family pairwise claims.

5.5 Same-family accounting

Table 16 reports the primary same-family MMLU-Pro accounting after the quality gate. These are within-archive accuracies under the paper's scoring protocol and should not be read as official benchmark scores. The results are useful as an archived-output case study, but they remain parser-dependent until the manual parser audit is completed.

Accuracy-improving directed comparisons still show substantial churn. Qwen has 20 accuracy-improving directed comparisons with mean churn of 35.2%, mean improvement mass of 26.7%, and mean regression mass of

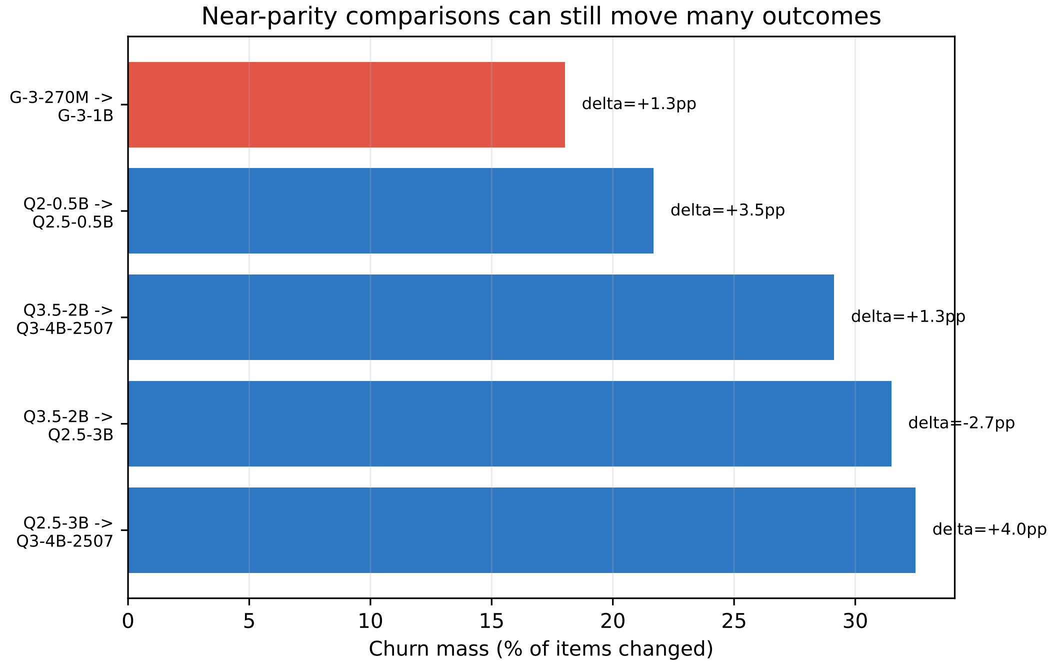

Fig. 1. All primary near-parity comparisons, defined as absolute accuracy delta at most five percentage points. Small aggregate movement can still correspond to large item-level churn, but path interpretation determines whether the comparison is a plausible deployment replacement or only an all-pairs diagnostic.

8.5%. Gemma has six accuracy-improving directed comparisons with mean churn of 27.3%, mean improvement mass of 19.5%, and mean regression mass of 7.9%.

Within this archived diagnostic, Qwen comparisons have lower average old-error persistence than Gemma comparisons. The contrast is not confirmatory because the archive is selected, parser-dependent, and incompletely manifested. The dependence-aware permutation check and model-level diagnostic figure are therefore reported only in the appendix.

5.6 Near-parity diagnostics and domain triage

Near-parity directed diagnostics are especially important because small aggregate movement can hide large item-level changes. Five primary MMLU-Pro comparisons have absolute accuracy delta at most five percentage points. Four are accuracy-improving directed diagnostics, and one is a negative-delta near-parity swap rather than an upgrade candidate model. Across these cases, small aggregate movement still corresponds to 18.0–32.5% item-level churn.

Domain summaries are used only to identify where manual review should start. They are not hypothesis tests and do not support causal domain-by-family claims.

Family	Category	Pairs	Δ	I	R
Gemma	health	6	10.0	21.1	11.1
Gemma	history	6	5.2	16.3	11.1
Gemma	computer science	6	9.9	20.0	10.1
Qwen	history	21	9.2	21.1	11.8
Qwen	law	21	4.2	15.3	11.1
Qwen	computer science	21	17.1	27.7	10.6

Table 17. Exploratory categories with the highest mean regression mass within each primary family. Values are percentages over category items averaged across eligible same-family comparisons.

Source	Tier	Access	Paper use
WILD item-level correctness	B	public	core item-level evidence
WILD raw records	A	public_large	not redistributed
Open LLM Leaderboard aggregates	C	public	context only
Open LLM Leaderboard details	D	gated_terms_required	excluded unless access granted
OpenCompass predictions	D	gated_terms_required	excluded unless access granted
Model cards and reports	C	public_or_model-gated	context only

Table 18. External evidence map used by the artifact. Only Tier A/B item-level sources can support replacement-accounting metrics. Tier C sources are context only, and Tier D sources are logged as exclusions unless access is later granted.

6 External Evidence Availability

The artifact classifies external evidence by what it can prove. This distinction is not bookkeeping; it is the boundary between valid replacement accounting and unsupported extrapolation. Error persistence, regression mass, and churn require paired item-level outcomes. Aggregate leaderboards can contextualize model quality, but they cannot identify corrected errors or newly introduced regressions.

Public model cards and technical reports provide useful aggregate context for small-model capability (Gemma Team 2024, 2025; Grattafiori et al. 2024; Qwen Team 2024; A. Yang et al. 2025), but they are not substitutes for paired item-level outputs. The artifact records source-backed aggregate scores and source pointers as context-only evidence in `analysis/external_benchmark_context/`. These scores may differ in prompt format, shot count, model variant, benchmark revision, and evaluation harness. For that reason, no paper claim about error persistence, regression mass, or churn is derived from external aggregate scores.

7 Discussion

7.1 What the evidence supports

The public WILD demonstration supports the core audit claim without answer extraction: aggregate gains can coincide with large item-level churn. The MMLU-Pro archived-output case study supports the same practical message under a raw-output workflow, while also demonstrating why source manifests, parser audits, scoring-mode sensitivity, and run manifests matter.

The MMLU-Pro family result should be read narrowly. In the evaluated archived runs, Qwen comparisons show lower old-error persistence than Gemma comparisons. The evidence does not establish a generalizable property of either model family.

7.2 What the evidence does not support

The evidence does not support a universal scaling law, a mechanistic explanation of why families differ, or a claim that one family is generally better for deployment. It also does not support interface-tax, long-context, safety, tool-use, multimodal, or multi-turn claims. The WILD results are correctness-only. The MMLU-Pro results are parser-dependent until the manual audit is complete.

7.3 How practitioners should use the audit

For an actual deployment, the audit should be run on the proposed successor path and on the task distribution that matters for the product. A high-churn replacement may be acceptable when corrected items are more important than regressions. That is a product and risk decision, not a conclusion implied by aggregate accuracy.

The audit card should be treated as a triage tool. Large regression mass, high normalized regression burden, and near-parity churn should trigger manual inspection of affected item clusters before deployment.

8 Threats to Validity

Construct validity. The audit measures changes in binary correctness over shared labeled items. It does not directly measure usefulness, calibration, safety, open-ended quality, or user satisfaction. WILD avoids parser error but is limited to correctness records. MMLU-Pro uses raw outputs but depends on deterministic answer extraction.

Parser validity. The MMLU-Pro parser may misclassify some raw completions, especially fallback cases. The generated 452-row manual audit sample includes 120 high-risk rows, but it must still be human-labeled before the raw-output case study can be treated as a fully validated model-evaluation result. This is the primary blocker for treating the MMLU-Pro case study as validated model evaluation. Until then, MMLU-Pro churn and regression claims are correctness-under-parser claims. The saved-prediction sensitivity check shows that scoring mode can change per-row correctness for runs with both saved predictions and raw responses.

Internal validity. The archived MMLU-Pro run logs do not preserve exact repository identifiers, model revision hashes, prompt templates, chat templates, decoding parameters, runtime versions, run dates, or complete hardware metadata. The reported accounting can be regenerated from saved outputs, but model-behavior replication requires full run manifests or reruns.

External validity. The primary raw-output case study uses one fixed MMLU-Pro subset. WILD adds public cross-benchmark correctness evidence over a selected subset of models and tasks, but it does not include the same prompts, raw completions, or parser decisions. Additional raw-output datasets and model families are needed for stronger generalization.

Selection and direction. The MMLU-Pro all-pairs diagnostic uses a deterministic ordering rule, not a complete vendor release chronology. Directed comparisons marked as chronology unclear should not be interpreted as real deployment replacements.

9 Artifact Outputs and Reproduction

The artifact is organized around generated, inspectable files rather than hand-computed paper values. The MMLU-Pro pipeline emits pairwise metrics as CSV/JSON, replacement-audit cards as CSV/JSON/Markdown, quality-gate summaries, parser-audit summaries, scoring-mode sensitivity tables, source-manifest hashes, and paper-ready LaTeX tables. The WILD pipeline emits normalized correctness records, pairwise replacement metrics, task-family summaries, task-dispersion summaries, a source-coverage report, and paper-ready tables. The external-evidence

Field	Value
Repository URL	https://github.com/DsChauhan08/error_persistence_regression_gap
Code license	MIT
Data-output license	CC-BY-4.0 for derived aggregate outputs; source datasets remain under their original terms
Python / OS	3.14.5 / Linux-7.0.10-101.fc43.x86_64-x86_64-with-glibc2.42
Dependency file	requirements.txt
Expected runtime	analysis regeneration: minutes; WILD ingest depends on download/cache; CPU rerun optional
Latest test command	PYTHONPATH=src pytest -q -p no:cacheprovider tests
Latest test result	43 passed in 4.24s
WILD item-level gate	true
MMLU-Pro source manifest	true
Public hygiene scan passed	true
Public manifest verified	true
Core WILD-based claim supported	true
Parser validation complete	false
MMLU-Pro confirmatory	false
GitHub/SSRN package ready	true
Blockers	parser_validated=false; non-blocking for GitHub/SSRN because MMLU-Pro raw-output claims are diagnostic only; mmlu_pro_confirmatory=false; non-blocking for GitHub/SSRN because WILD carries the core claim

Table 19. Artifact release-status gate for the GitHub/SSRN public package. Parser validation and MMLU-Pro confirmatory status remain visible but do not block the parser-independent WILD/source-manifest claim set.

pipeline emits a source-quality map, aggregate context tables, informal-context exclusions, and a claim-use manifest.

The artifact also includes a CPU-only mini-rerun harness for fully manifested follow-up validation. It writes a run manifest, environment file, selected-item hash manifest, resumable raw records, per-model summaries, pairwise audit metrics, and claim checks. A completed real-model mini-rerun is not used as evidence in the present manuscript unless its outputs, exact model revisions, prompt template, tokenizer/chat-template metadata, decoding settings, runtime environment, hardware summary, and run date are archived with the release.

The release also includes tests for source-manifest matching, parser-audit summaries, scoring-mode sensitivity, WILD replacement accounting, evidence-tier restrictions, CPU-rerun resume behavior, and public-release hygiene. A hygiene scan checks that public artifacts do not contain raw model responses, benchmark text, local cache paths, or private files.

10 Data and Software Availability

The GitHub/SSRN public proof-of-analysis package is prepared for release at https://github.com/DsChauhan08/error_persistence_regression_gap. The package contains the manuscript, analysis code, derived MMLU-Pro pairwise metrics, source-row hashes, parser-audit metadata, scoring-mode sensitivity summaries, WILD correctness-only replication outputs, evidence-tier manifests, generated tables, and generated figures. It does not redistribute full MMLU-Pro question text, full answer-option text, raw model responses, or item-level response excerpts.

The accounting can be regenerated from saved outputs, but full model-behavior replication requires complete run manifests. A future rerun intended to support confirmatory model-evaluation claims should archive exact model repository IDs, revision hashes, prompt templates, chat templates, decoding parameters, inference engine, package versions, hardware summary, and run dates. The absence of completed manual parser labels is therefore not hidden or bypassed: parser-dependent MMLU-Pro claims are excluded from the core claim set in Table 2.

The main public-package reproduction commands are:

```
PYTHONPATH=src python -m boundary_slm.paper_metrics
PYTHONPATH=src python -m boundary_slm.mmlu_pro_manifest
PYTHONPATH=src python -m boundary_slm.external_evidence_registry
PYTHONPATH=src python -m boundary_slm.external_wild_ingest
PYTHONPATH=src python -m boundary_slm.artifact_release_status \
  --repository-url https://github.com/DsChauhan08/error_persistence_regression_gap
PYTHONPATH=src python -m boundary_slm.artifact_manifest
PYTHONPATH=src python -m boundary_slm.public_release_hygiene
PYTHONPATH=src pytest -q
```

Parser-audit regeneration and scoring-mode sensitivity over private/source raw outputs require access to the withheld raw model-output JSONL files:

```
PYTHONPATH=src python -m boundary_slm.scoring_mode_sensitivity
PYTHONPATH=src python -m boundary_slm.parser_audit_impact
PYTHONPATH=src python -m boundary_slm.parser_audit_second_pass
PYTHONPATH=src python -m boundary_slm.mmlu_scoring_robustness
```

The public package keeps only redacted audit metadata and aggregate summaries.

11 Conclusion

This paper presents an item-level audit for small language model replacement. Aggregate accuracy says whether a candidate model wins on average. The proposed decomposition shows how it wins, what it fails to fix, and what it breaks.

The public WILD demonstration shows that the audit can be applied to independent item-level correctness data and that modest aggregate gains can still involve large churn. The MMLU-Pro archived-output case study shows the same practical issue in a raw-output workflow, while also exposing the need for source manifests, parser validation, scoring-mode sensitivity, and full run provenance.

The practical recommendation is simple: benchmark-based model replacement decisions should report corrected errors, regressions, persistent errors, and churn mass, not only aggregate accuracy.

12 Author Contributions

Dhananjay Singh Chauhan designed the study, ran or assembled the model-output evaluations, implemented the analysis code, performed the statistical analysis, generated the tables and figures, and wrote the manuscript.

13 Competing Interests

The author declares that there are no competing interests.

14 Grant Information

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A MMLU-Pro Model Runs and Quality Gate

Model label	Family	Params (B)	Accuracy	Answered	Primary role
Qwen3-4B-Instruct-2507	Qwen	4.00	48.2	100.0	paired
Qwen3.5-2B	Qwen	2.00	46.9	99.8	paired
Qwen2.5-3B-Instruct	Qwen	3.00	44.2	99.7	paired
Qwen3.5-0.8B	Qwen	0.80	33.0	99.6	paired
Gemma-2-2B-Instruct	Gemma	2.00	31.7	99.4	paired
Qwen3-0.6B	Qwen	0.60	28.0	100.0	paired
Llama-3.2-1B-Instruct	Llama	1.00	21.2	98.2	anchor
Gemma-3-1B-it	Gemma	1.00	17.5	98.1	paired
Qwen2.5-0.5B-Instruct	Qwen	0.50	16.5	99.3	paired
Qwen2-0.5B-Instruct	Qwen	0.50	13.0	98.7	paired
Phi-2	Phi	2.70	11.8	90.8	anchor
Gemma-3-1B-Instruct	Gemma	1.00	11.8	99.1	paired
Gemma-3-270M-Instruct	Gemma	0.27	10.4	95.5	paired
Gemma-4-E2B-Instruct	Gemma	2.00	1.8	5.2	excluded

Table 20. Model-level leaderboard and quality-gate status for the archived MMLU-Pro outputs. Accuracy and answered values are percentages over the same 3,008-item set.

B MMLU-Pro Run Metadata Status

Metadata field	Captured for all models?	Models with field	Total models	Complete	Missing	Note
Item id	yes	14	14	14	0	captured
Category	yes	14	14	14	0	captured
Ground-truth label	yes	14	14	14	0	captured
Raw response	yes	14	14	14	0	captured
Saved prediction	no	3	14	3	11	missing/incomplete
GPU worker id	no	2	14	2	12	missing/incomplete
HF repository	no	0	14	0	14	missing/incomplete
Model revision hash	no	0	14	0	14	missing/incomplete
Prompt template	no	0	14	0	14	missing/incomplete
Chat template	no	0	14	0	14	missing/incomplete
Decoding parameters	no	0	14	0	14	missing/incomplete
Runtime/package versions	no	0	14	0	14	missing/incomplete
Run date	no	0	14	0	14	missing/incomplete

Table 21. Run-metadata completeness in the archived MMLU-Pro raw-output files.

C Diagnostic MMLU-Pro Family Check

For the archived MMLU-Pro diagnostic only, each eligible model is assigned a neighborhood value by averaging over its incident same-family directed pairs. An exact family-label permutation check over these model-level values gives lower old-error persistence for Qwen than Gemma in this archive ($p = 0.003$, Holm-adjusted $p = 0.006$ over two primary contrasts). Normalized regression burden is not a robust family separator ($p = 0.073$). These p-values quantify a contrast in a small selected archive; they are not confirmatory family-level inference because the model set is selected, parser-dependent, incompletely manifested, and not a random sample of releases.

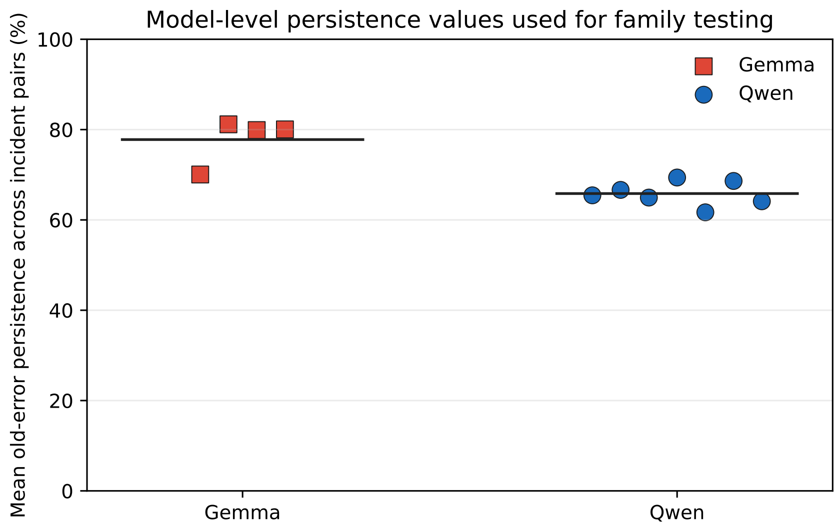

Fig. 2. Diagnostic MMLU-Pro persistence values with parser audit incomplete. Each point is one eligible model averaged over its incident same-family comparisons; horizontal bars mark family means.

D Primary MMLU-Pro Directed Comparisons

Family	Directed pair	Path note	Δ	*I*	*R*	Churn	Persist.	Corr.	NRB	Review
Qwen	Qwen2-0.5B-Instruct → Qwen2.5-0.5B-Instruct	newer	3.5	12.6	9.1	21.7	85.6	14.4	10.5	manual required
Qwen	Qwen2-0.5B-Instruct → Qwen3-0.6B	newer+larger	15.0	22.6	7.6	30.2	74.1	25.9	8.7	manual advised
Qwen	Qwen2.5-0.5B-Instruct → Qwen3-0.6B	newer+larger	11.5	20.8	9.3	30.2	75.0	25.0	11.1	manual required
Qwen	Qwen2-0.5B-Instruct → Qwen3.5-0.8B	newer+larger	20.0	27.1	7.1	34.2	68.8	31.2	8.1	manual advised
Qwen	Qwen2.5-0.5B-Instruct → Qwen3.5-0.8B	newer+larger	16.6	25.4	8.8	34.2	69.6	30.4	10.5	manual required
Qwen	Qwen2-0.5B-Instruct → Qwen3.5-2B	newer+larger	33.9	39.8	5.9	45.7	54.3	45.7	6.8	manual advised
Qwen	Qwen2.5-0.5B-Instruct → Qwen3.5-2B	newer+larger	30.5	36.5	6.1	42.6	56.3	43.7	7.2	manual advised
Qwen	Qwen2-0.5B-Instruct → Qwen2.5-3B-Instruct	newer+larger	31.2	37.2	6.0	43.1	57.3	42.7	6.8	manual advised
Qwen	Qwen2.5-0.5B-Instruct → Qwen2.5-3B-Instruct	same-gen larger	27.8	34.0	6.3	40.3	59.3	40.7	7.5	manual advised
Qwen	Qwen2-0.5B-Instruct → Qwen3-4B-Instruct-2507	newer+larger	35.2	41.2	5.9	47.1	52.7	47.3	6.8	manual advised
Qwen	Qwen2.5-0.5B-Instruct → Qwen3-4B-Instruct-2507	newer+larger	31.8	38.3	6.5	44.9	54.1	45.9	7.8	manual advised
Qwen	Qwen3-0.6B → Qwen3.5-0.8B	newer+larger	5.1	17.0	11.9	28.9	76.5	23.5	16.5	manual required
Qwen	Qwen3-0.6B → Qwen3.5-2B	newer+larger	18.9	26.6	7.6	34.2	63.1	36.9	10.6	manual required
Qwen	Qwen3-0.6B → Qwen2.5-3B-Instruct	larger; chronology unclear	16.2	24.7	8.5	33.2	65.7	34.3	11.8	manual required
Qwen	Qwen3-0.6B → Qwen3-4B-Instruct-2507	same-gen larger	20.2	27.4	7.1	34.5	62.0	38.0	9.9	manual advised
Qwen	Qwen3.5-0.8B → Qwen3.5-2B	same-gen larger	13.9	22.2	8.3	30.5	66.9	33.1	12.4	manual required
Qwen	Qwen3.5-0.8B → Qwen2.5-3B-Instruct	larger; chronology unclear	11.2	21.9	10.7	32.6	67.3	32.7	16.0	manual required
Qwen	Qwen3.5-0.8B → Qwen3-4B-Instruct-2507	larger; chronology unclear	15.2	25.0	9.8	34.8	62.7	37.3	14.6	manual required
Qwen	Qwen3.5-2B → Qwen2.5-3B-Instruct	negative swap	-2.7	14.4	17.1	31.5	72.9	27.1	32.2	not upgrade
Qwen	Qwen3.5-2B → Qwen3-4B-Instruct-2507	larger; chronology unclear	1.3	15.2	13.9	29.1	71.3	28.7	26.2	manual required
Qwen	Qwen2.5-3B-Instruct → Qwen3-4B-Instruct-2507	newer+larger	4.0	18.3	14.2	32.5	67.3	32.7	25.5	manual required

Table 22. Primary Qwen MMLU-Pro directed comparisons after the answerability gate. Values are percentages.

Family	Directed pair	Path note	Δ	I	R	Churn	Persist.	Corr.	NRB	Review
Gemma	Gemma-3-270M-Instruct ⟶ Gemma-3-1B-Instruct	same-gen larger	1.3	9.7	8.3	18.0	89.2	10.8	9.3	manual required
Gemma	Gemma-3-270M-Instruct ⟶ Gemma-3-1B-it	same-gen larger	7.1	15.6	8.5	24.0	82.6	17.4	9.5	manual advised
Gemma	Gemma-3-270M-Instruct ⟶ Gemma-2-2B-Instruct	larger; chronology unclear	21.2	28.4	7.1	35.5	68.3	31.7	8.0	manual advised
Gemma	Gemma-3-1B-Instruct ⟶ Gemma-3-1B-it	same-size variant	5.8	13.5	7.7	21.2	84.7	15.3	8.7	routine
Gemma	Gemma-3-1B-Instruct ⟶ Gemma-2-2B-Instruct	larger; chronology unclear	19.9	26.9	6.9	33.8	69.6	30.4	7.9	manual advised
Gemma	Gemma-3-1B-it ⟶ Gemma-2-2B-Instruct	larger; chronology unclear	14.2	22.8	8.7	31.5	72.3	27.7	10.5	manual required

Table 23. Primary Gemma MMLU-Pro directed comparisons after the answerability gate. Values are percentages.

E Descriptive Gain–Churn Geometry

Family	Metric	Pairs	Pairwise r	95% CI	LOO range
Gemma	Norm. burden	6	-0.472	[-0.928, 0.550]	[-0.992, -0.105]
Gemma	Churn mass	6	0.988	[0.894, 0.999]	[0.963, 1.000]
Gemma	Persistence	6	-0.988	[-0.999, -0.891]	[-1.000, -0.969]
Qwen	Norm. burden	21	-0.811	[-0.920, -0.583]	[-0.891, -0.727]
Qwen	Churn mass	21	0.897	[0.759, 0.958]	[0.874, 0.951]
Qwen	Persistence	21	-0.856	[-0.940, -0.673]	[-0.898, -0.861]

Table 24. Descriptive correlations between accuracy delta and item-level metrics. These are not independent-observation tests because model pairs share endpoints.

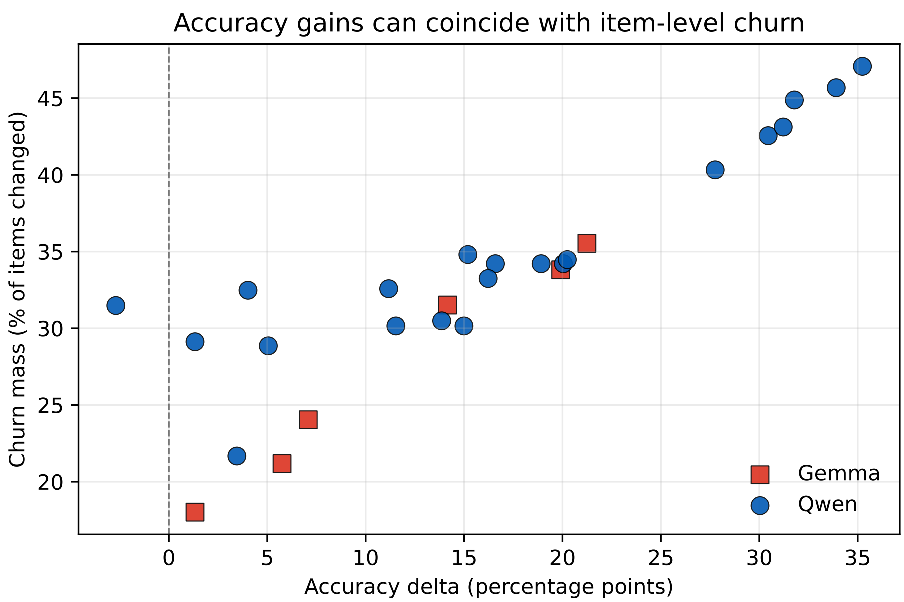

Fig. 3. Accuracy delta versus churn mass for primary same-family MMLU-Pro comparisons. The scatter is descriptive: both axes are functions of the same improvement and regression counts.

F Answer-Extraction Rules

Answer extraction is deterministic. If a row contains a saved `prediction` field equal to one valid option letter, that field is used. Otherwise the parser searches the raw response in the following order and returns the last matching valid option letter:

(1) A single-letter response, optionally wrapped in parentheses or followed by a period.
(2) Explicit final-answer markers: `final answer`, `correct answer`, `correct option`, `correct choice`, `therefore`, `answer is`, and `option is`.

(3) Tail option markers in the final 500 characters of the response, such as a standalone option letter followed by punctuation.
(4) The final standalone option letter in the response, used only as a low-confidence fallback.

Before these rules are applied, a prompt-echo guard marks a response as unanswered if it ends in a bare chat marker after echoed prompt text and contains no explicit answer marker. The valid option alphabet is `A–J`, matching the saved ground-truth labels. Rows with no extracted valid option are counted as unanswered and incorrect.